

Problem set 5

Clustering

Exercise 1

Compute the confusion matrix, macro-averaged Precision, Recall, and F-score for the clustering shown below.

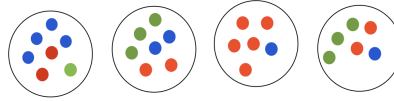


Figure 1: Caption

Solution First cluster will be labelled as blue. Second cluster will be labelled as green. Third cluster will be labelled as red. Fourth cluster will be labelled as green.

Hence Second and fourth cluster will be combined.

Confusion matrix is given below.

$$\begin{pmatrix} & \text{blue} & \text{red} & \text{green} \\ \text{blue} & 4 & 1 & 3 \\ \text{red} & 2 & 5 & 4 \\ \text{green} & 1 & 0 & 6 \end{pmatrix}$$

Now for blue labelled cluster precision is $\frac{4}{7}$. For green labelled cluster, the precision is $\frac{6}{13}$. For red labelled cluster the precision is $\frac{5}{6}$. Hence macro averaged precision is $(0.571 + 0.461 + 0.833)/3 = 0.621$

Now for blue class label recall is $\frac{4}{8}$. For green label, the recall is $\frac{6}{7}$. For red label, the recall is $\frac{5}{11}$. Hence macro averaged recall is $(0.5 + 0.857 + 0.454)/3 = 0.603$

Macro averaged FScore is $(0.533 + 0.599 + 0.587)/3 = 0.573$

Exercise 2

For the same clusters as in Exercise 1, compute B-CUBED Precision, Recall, and F-score.

$$\text{BCubed Precision} = \frac{1}{26} \sum_{i=1}^{26} (4 * \frac{4}{7} + 2 * \frac{2}{7} + \frac{1}{7} + 6 * \frac{6}{13} + 3 * \frac{3}{13} + 4 * \frac{4}{13} + 5 * \frac{5}{6} + \frac{1}{6}) = 0.462$$

The recall and FScore can be computed in a similar way.

Excercise 3

Solution The euclidean distance between any pair of points is given below.

$$\begin{pmatrix} 0 & 5 & 8.49 & 3.61 & 7.07 & 7.21 & 8.06 & 2.24 \\ 5 & 0 & 6.08 & 4.24 & 5 & 4.12 & 3.16 & 4.47 \\ 8.49 & 6.08 & 0 & 5 & 1.41 & 2 & 7.28 & 6.4 \\ 3.61 & 4.24 & 5 & 0 & 1.41 & 4.12 & 5.39 & 5.39 \\ 7.07 & 4.12 & 2 & 1.41 & 0 & 5.39 & 7.28 & 6.4 \\ 7.21 & 3.16 & 7.28 & 5.39 & 7.21 & 0 & 5.39 & 7.62 \\ 8.06 & 4.47 & 6.4 & 5.39 & 6.4 & 7.62 & 0 & 2.24 \\ 2.24 & 4.47 & 6.4 & 5.39 & 6.4 & 7.62 & 2.24 & 0 \end{pmatrix}.$$

- First Step of the clustering $\{X_1\}, \{X_2\}, \{X_3, X_5\}, \{X_4\}, \{X_6\}, \{X_7\}, \{X_8\}$
- Second Step of the clustering $\{X_1\}, \{X_2\}, \{X_3, X_5\}, \{X_6\}, \{X_7\}, \{X_4, X_8\}$
- Third Step of the clustering $\{X_1\}, \{X_2\}, \{X_3, X_5, X_6\}, \{X_7\}, \{X_4, X_8\}$
- Fourth Step of clustering $\{X_1, X_4, X_8\}, \{X_2\}, \{X_3, X_5, X_6\}, \{X_7\}$
- Fifth Step of clustering $\{X_1, X_4, X_8\}, \{X_2, X_7\}, \{X_3, X_5, X_6\}$
- Sixth Step of clustering $\{X_1, X_4, X_8, X_3, X_5, X_6\}, \{X_2, X_7\}$
- Seventh Step of clustering $\{X_1, X_4, X_8, X_3, X_5, X_6, X_2, X_7\}$

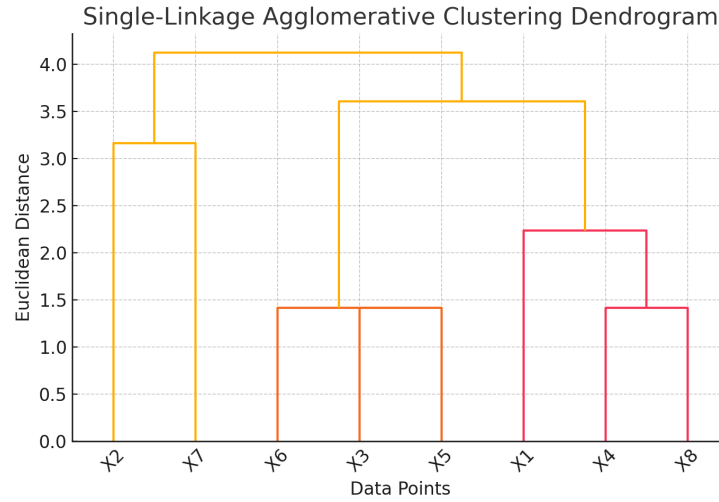


Figure 2:

Exercise 4

The dataset consists of four objects with pairwise distances given by the following distance matrix

$$\begin{pmatrix} & X_1 & X_2 & X_3 & X_4 \\ X_1 & 0 & 2 & 6 & 2.5 \\ X_2 & 2 & 0 & 3 & 8 \\ X_3 & 6 & 3 & 0 & 4 \\ X_4 & 2.5 & 8 & 4 & 0 \end{pmatrix}.$$

1. Cluster dataset using single-linkage agglomerative clustering. Draw the corresponding dendrogram.
2. Cluster dataset using complete-linkage agglomerative clustering. Draw the corresponding dendrogram.

Solution First Step of Single Linkage Clustering $\{X_1, X_2\}, \{X_3\}, \{X_4\}$
 Second Step of Single Linkage Clustering $\{X_1, X_2, X_4\}, \{X_3\}$
 Third Step of Single Linkage Clustering $\{X_1, X_2, X_4, X_3\}$

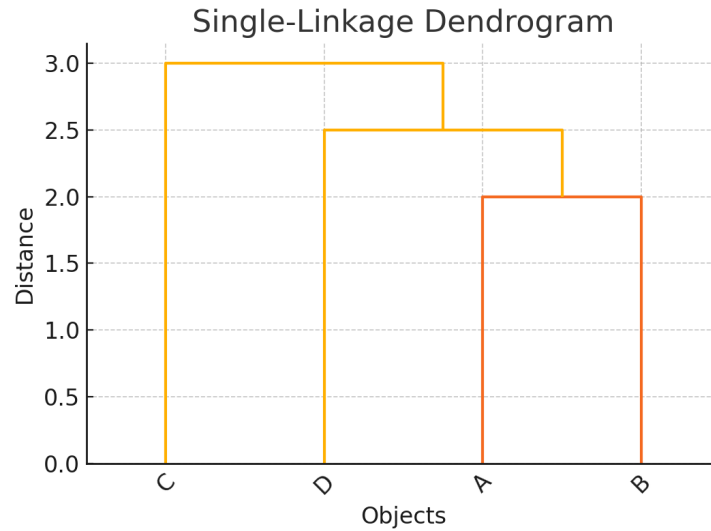


Figure 3:

First Step of Complete Linkage Clustering $\{X_1, X_2\}, \{X_3\}, \{X_4\}$
 Second Step of Complete Linkage Clustering $\{X_1, X_2\}, \{X_3, X_4\}$
 Third Step of Complete Linkage Clustering $\{X_1, X_2, X_4, X_3\}$

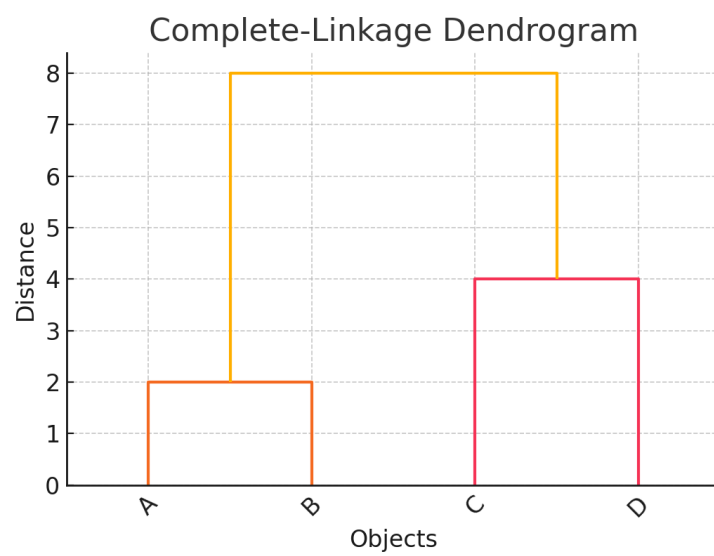


Figure 4: